IoT Panel Troubleshooting Guide

Overview

The AgenticIoT demo panel is a physical hardware panel connected to a Raspberry Pi. It consists of four switches and four LEDs wired to GPIO pins. The panel streams telemetry to Azure IoT Hub, which flows through Event Hub to an Azure Function, SignalR Service, and Power Pages dashboard in real time.

The Copilot Studio agent monitors incoming telemetry and assists users in diagnosing and resolving panel state mismatches.

Panel Hardware

Switches

Switch	Label	Physical Type	How to Operate
SW1	Switch 1	Military-style red toggle switch	Flip the toggle up or down to activate/deactivate
SW2	Switch 2	Military-style red toggle switch	Flip the toggle up or down to activate/deactivate
SW3	Switch 3	Keyed switch	Insert the physical key and turn to activate
SW4	Switch 4	Keyed switch	Insert the physical key and turn to activate

SW1 and SW2 are military-style red toggle switches used for standard on/off control. SW3 and SW4 are keyed switches that require a physical key to operate, providing an additional layer of access control.

LEDs

LED	GPIO (BCM)	Physical Pin	Colour
LED0	GPIO 18	Pin 12	Blue
LED1	GPIO 24	Pin 18	Orange
LED2	GPIO 25	Pin 22	Green
LED3	GPIO 12	Pin 32	Yellow

All LEDs are active-HIGH and wired with a 330-ohm series resistor (approximately 4 mA at 3.3 V).

GPIO Switch Wiring

Switch	GPIO (BCM)	Physical Pin	Logic
SW1	GPIO 5	Pin 29	Pull-up, LOW when active
SW2	GPIO 6	Pin 31	Pull-up, LOW when active
SW3	GPIO 13	Pin 33	Pull-up, LOW when active
SW4	GPIO 19	Pin 35	Pull-up, LOW when active

Logic Map

The panel applies switch-to-LED rules defined in `logic_map.json`. Rules are evaluated in order; the first match wins. If no rule matches, the fallback applies.

Rule ID	Switches Active	LEDs On	Description
`all_lights_on`	SW1 + SW3	LED0, LED1, LED2, LED3	Target healthy state — all 4 LEDs on
`single_light`	SW2 only	LED2	Green LED only
`pattern_lights`	SW1 + SW2	LED0, LED2	Blue + green
`diagnostic_mode`	SW1 + SW2 + SW3	LED1, LED3	Orange + yellow
`shutdown_sequence`	SW1 + SW2 + SW4	None	All LEDs off
`switch_4_only`	SW4 only	LED3	Yellow LED only
fallback	(no rule matched)	LED0	Default: blue LED only

Healthy State Definition

A panel is considered healthy when **all four LEDs (LED0, LED1, LED2, LED3) are ON simultaneously**.

The only switch combination that produces this state is **SW1 + SW3 active** (rule: `all_lights_on`).

This is always the target state regardless of the current mismatch, active rule, or fault flag present in telemetry.

How to Fix a Broken Panel

This section provides step-by-step instructions to restore the panel to its healthy state (all 4 LEDs on).

Goal

Get all four LEDs on at the same time. The only way to achieve this is to have SW1 and SW3 active simultaneously, with SW2 and SW4 inactive.

Step-by-Step Fix

1. **Locate SW3 (keyed switch)** — this is a keyed switch. You will need the physical key.
2. **Insert the key into SW3** and turn it to the active position.
3. **Locate SW1 (red toggle switch)** — flip the toggle to the active (up) position.
4. **Verify SW2 is inactive** — SW2 should be in the off/down position.
5. **Verify SW4 is inactive** — SW4 key should be turned to the off position (or key removed).
6. **Confirm the result** — all four LEDs (blue, orange, green, yellow) should now be lit. If not, check the telemetry on the dashboard and repeat from step 1.

Common Mismatch Scenarios

Observed State	Likely Cause	Fix
Only LED0 (blue) on	No rule matched — SW combination not recognised	Activate SW1 + SW3, deactivate SW2 and SW4

Only LED2 (green) on	SW2 only active (rule: `single_light`)	Activate SW1 and SW3; deactivate SW2
LED0 + LED2 on	SW1 + SW2 active (rule: `pattern_lights`)	Activate SW3; deactivate SW2
LED1 + LED3 on	SW1 + SW2 + SW3 active (rule: `diagnostic_mode`)	Deactivate SW2; keep SW1 + SW3 active
All LEDs off	SW1 + SW2 + SW4 active (rule: `shutdown_sequence`)	Deactivate SW2 and SW4; keep SW1; activate SW3
Only LED3 (yellow) on	SW4 only active (rule: `switch_4_only`)	Deactivate SW4; activate SW1 + SW3

Quick-Fix Summary (plain steps)

When a user reports the panel is broken or requests a quick fix, provide only these steps:

1. Turn SW3 key to the active position.
2. Flip SW1 toggle to the active (up) position.
3. Ensure SW2 toggle is in the off (down) position.
4. Ensure SW4 key is in the off position.
5. Confirm all four LEDs are now lit.

Telemetry and `needs_help` Flag

When the panel detects a mismatch between expected and actual state, it sets `needs_help: true` in the telemetry payload sent to Azure IoT Hub. This flag triggers the Copilot agent to start a diagnostic session.

The telemetry payload also includes:

- `switches` : array of active switch indices (1-based)
- `leds` : array of active LED indices (0-based)
- `active_rule` : the rule ID currently matched, or `null` if no rule matched
- `needs_help` : boolean mismatch flag

Azure Resource Names

Resource	Name
IoT Hub	`iothub-aw-iot-copilot`
Event Hub Namespace	`evhns-aw-iot-copilot`
Event Hub	`iot-telemetry`
Azure Function App	`func-aw-iot-copilot`
SignalR Service	(configured in Function App settings)
Resource Group	`rg-aw-azcom-iot-copilot`
IoT Device ID	`raspberry-pi-iotpanel`

Troubleshooting Common Issues

Panel not streaming telemetry

1. Verify the Raspberry Pi is powered on and the `iot-monitor` service is running: `sudo systemctl status iot-monitor`
2. Check the IoT Hub connection string is set in `/opt/iot-monitor/.env`.
3. Verify network connectivity from the Pi to Azure.
4. Check IoT Hub routing in the Azure Portal — confirm the route to the `iot-telemetry` Event Hub is enabled.

Dashboard not updating in real time

1. Confirm the Azure Function App (`func-aw-iot-copilot`) is running.
2. Check the `IoTHubEventHubConnectionString` app setting is populated in the Function App.
3. Verify the SignalR Service connection in the Function App configuration.
4. Reload the Power Pages dashboard and check browser console for SignalR connection errors.

LED does not respond to switch

1. Check the GPIO wiring against the pin table in the Panel Hardware section above.
2. Verify the switch pull-up is working — measure pin voltage when switch is open (should be ~3.3 V) and when closed (should be ~0 V).
3. Check `panel_controller.py` for the correct BCM pin numbers.
4. Restart the `iot-monitor` service on the Pi.